SWA2124 FINAL ASSESSMENT

Liang Yrenn Da (22067128), Yeoh Jie Long (22071112), Howell Tek Teck Kee (22055479) , Ooi Jin Yon (22039069)

Abstract— This research paper explores how to predict the performance of secondary school students in mathematics using an upstream student performance dataset raised by UCI. The study aims to determine the relative impact of personal or external variables on the final grade of students in mathematics (G3). Although there is growing interest in data analytics related to education, their relative significance as factor categories is uncertain. The goal is to compare predictive power of individual characteristics (previous grades and attendance) and external dimensions (parent education and school support), as well as assess the precision of a variety of machine learning models. The four models, including linear regression, decision tree, random forest, and support vector machine, were trained and tested with the standard error and accuracy as the metrics. The decision tree model obtained the best predictive performance ($R^2 = 0.91$), and feature importance analysis indicated that the most significant influence on academic performance was the effect of personal factors as compared to external variables. These lessons can guide teachers to take priority over students at risk and focus on early intervention based on their learning history and behaviours.

Keywords— *student performance, predictive analytics, machine learning, decision tree, education data mining*

I. Introduction

It is pertinent to accurately forecast the academic performance of students as education institutions aim at offering timely academic support and enhancing the results of education. Students who are at-risk of underperforming can be identified early on in their education so that targeted interventions, including tutoring, counselling, or tailored learning plans, can be implemented, which can have a dramatic effect on student success and retention. With the growing volume and availability of educational data, there is a growing demand to use advanced analytics / machine learning to support decision-making in the academic field.

The academic performance of the students involves a complicated combination of personal and external variables. However, many studies have observed that personal variables, including previous grades, study habits, motivation, and attendance as well as a history of academic failures are good predictors of future academic achievement. Simultaneously, family socioeconomic status (SES), parental involvement, as well as school environment are external factors that are generally recognized to exert an effect on educational results. In another example, it has been found that parental involvement has more significant impacts on students with more socioeconomic status [1]. Researchers in other studies argue that school related elements like classroom resources and infrastructure are very crucial in determining student achievement [2]. Furthermore, it has also been highlighted that family SES and the level of academic involvement shown by the parents equally contribute to academic achievement [3]. Although these results exist, the relative predictive strength of these classifications personal versus external has not been sufficiently developed especially within data-driven modelling.

To counter this, increasingly machine learning and statistical modelling have been used to predict student performance. Methods such as logistic regression, decision trees, random forests, and support vector machines are commonly used to model and predict grades within an academic environment, and each algorithm has its distinct benefits, related to its accuracy as well as its interpretability. A research study was conducted to prove that the application of a combination of machine learning algorithms can be used successfully to establish notable performance predictors in students [9]. The verification of the second study affirmed that logistic regression yields useful information on the variables that impact on the mathematics grades [12]. Nevertheless, a typical constraint exists in previously conducted studies, either by concentrating on a small group of features or a single model learning strategy. As a result, the literature continues to lack consensus on the comparative effect of both personal and external characteristics on academic performance within the backdrop of sound predictive analytics models.

This research seeks to fill these gaps by engaging in a comparative study of the predictability of personal and external variables to the final grade of the students of the mathematics subject (G3). In particular, we consider a publicly released data of 395 secondary school children across two Portuguese schools and use four regression algorithms, including linear regression, decision tree, random forest, and support vector machine regression, to predict final performance. Our procedure consists of a three-step feature selection method (correlation analysis, p-value filter, and random forest importance ranking) to find most relevant predictors and then model training and evaluation with standard measures (MAE, MSE, RMSE, R 2). The critical aspect is to quantify which subset of factors has the greater impact on predictive accuracy and what sort of modelling is the most effective in this educational scenario.

This study is restricted to predicting final mathematics grades based on a UCI Student Performance dataset, including demographic, personal, and family related data, and history of previous academia. All data relate only to grades in maths and their characteristic features; no other subjects or datasets are considered. The study fails to capture the effects of the unmeasured factors, including psychological well-being or peer influence, and does not take into consideration the longitudinal changes in performance besides the accessible data.

The contributions of this study are that it can shape both policy and practice in education. This study helps educators, administrators, and policymakers to apply evidence-based knowledge on their early intervention and student support resources by identifying which types of factors can effectively predict academic outcomes based on empirical

analysis. Furthermore, the orderly evaluation of several modelling methods, along with the formalized feature selection procedure, provides methodological contribution to research on both educational data mining and analytics in learning.

These contributions are threefold: (1) an empirical comparison of four regression methods in predicting grades, (2) an elaborate study of the relative significance of personal and external factors influencing performance, and (3) practical insight that can inform data-driven interventions in the educational setting. This paper is structured as follows: in Section II, literature that is relevant will be reviewed, in Section III the methodology, including data collection, feature selection, and model evaluation will be described, in Section IV the results will be presented and discussed, and finally in Section V, this paper will be concluded with implications and future research suggestions.

II. Literature Review

A. Structure of the Literature Review

This literature review focus around four key concepts that examine the various dimensions impacting students' academic achievement. The first section focuses on external factors, including family socioeconomic status, parental involvement and school environmental conditions. These articles show the impact of external support resources that students gain beyond their personal control [1][2][3]. The second section internal factors, such as motivation, learning strategies and time management. These articles highlight how students' own attitudes, behaviours, and cognitive strategies play an important role in students' academic achievement [4][5][6]. The third section examines the use of matrix frameworks to analyse or categorize factors influencing academic achievement. These articles give structured analytical models that are helpful in looking at how a number of variables interact. [7][8][9]. Finally, the fourth section discusses the use of theoretical models in analyzing academic performance, offering conceptual frameworks for looking at student success from a broader perspective. This structure allows for comparing comprehensively individual, environmental, and theoretical influences on academic success [10][11][12].

B. Writing the Literature Review

Reference	Title	Problem Statement	Technique	Findings	Limitations
[1]	Academic Benefits from Parental Involvement are Stratified by Parental Socioeconomic Status: A Meta-analysis	The study addresses whether parental involvement benefits students equally across socioeconomic backgrounds, questioning the assumption that all students gain similar academic advantages regardless of parental SES.	**Design:** Meta-analysis **Sample:** 98 quantitative studies from various SES backgrounds and grade levels **Method:** Pearson correlation coefficient to extract effective size and SES indicators for parental education **Analysis:** Fisher's z-transformation, weighted effect size calculation, moderation analysis by SES	The study found that subtle forms of parental involvement, such as academic expectations and reading together, had the strongest positive effects on student achievement, particularly among students from higher-SES families.	The study relied solely on parental education as the SES measure, which may not capture financial or social capital. It also assumed equal access to involvement resources among parents and only included quantitative studies, limiting depth of insight.
[2]	School environmental factors influencing academic performance in secondary schools	The study investigates how physical and environmental conditions in secondary schools such as classroom ventilation, noise, furniture, lighting, and instructional materials on affect students' academic performance in English Language in Nigeria.	**Design:** Descriptive survey research design **Sample:** 250 students from five randomly selected public secondary schools in Oyo State, Nigeria **Tool:** Structured questionnaire and achievement test in English Language **Analysis:** Descriptive statistics (frequency, mean), and inferential analysis using t-test and ANOVA	This study found that school environmental factors significantly related with students' academic performance in English Language. Exam scores were higher in classrooms that had adequate lighting, adequate ventilation, seating arrangements, and instructional materials. Furthermore, classroom environmental conditions with high levels of noise and overcrowding had a detrimental impact on students' concentration and performance.	While this study made programme recommendations based on classroom environmental conditions, only participants from public schools in one state in Nigeria were participants and it will only extend to English Language performance which will limit generalizability. Also, the participants were only relying self-reported questionnaire responses in conjunction with their test scores which could be affected by social factors that were not apparently captured in the study.

[3]	Family socio-economic status and children's academic achievement: The different roles of parental academic involvement and subjective social mobility	Examined the mediating role of parental academic involvement and the moderating role of parental subjective social mobility in this relationship with cross-sectional data	A questionnaire about demographic information and parental academic are completed by 966 sixth-grade students from five rural public elementary schools in northern and southern China The study examined direct and indirect effects of SES on student achievement, highlighting mediating roles of educational expenditure and parental expectations. **Design:** Cross-sectional quantitative study **Sample:** 1,387 junior high school students from the 2018 China Family Panel Studies (CFPS) dataset **Tool:** Structured national survey data on family SES, academic performance, parental expectations, and educational spending **Analysis:** Descriptive statistics, regression analysis, and mediation modelling using SPSS and PROCESS macro	Family SES had a significant positive impact on students' academic achievement. Parental expectations and educational expenditure both partially mediated this relationship, with higher SES families showing greater investment and higher expectations, leading to better academic outcomes.	Two indicators of family SES used in previous studies may not be comprehensive. Rather, three common indicators, namely parents' education, parents' occupation, and family income, may better reflect the general levels of family SES. Secondly, few studies have focused on parental academic involvement as reported by children. Evidence has indicated that the levels of parental academic involvement as self-reported by parents are usually higher than those reported by children
[4]	The impact of study habits and personal factors on the academic achievement performances of medical students	Medical students often face challenges in achieving high academic performance, influenced by both study habits and personal factors. Existing studies lack regional data, especially from Saudi Arabia, and there's limited understanding of which specific habits correlate with higher academic success among medical students	**Design**: Observational cross-sectional study **Sample**: 336 medical students (1st to 5th year) from King Saud University **Tool**: Self-developed electronic questionnaire (3 sections: demographics, personal factors, study habits) **Analysis:** Statistical analysis using SPSS, odds ratios with significance level at $p < 0.05$ Students were split into high GPA (≥4.5/5) and low GPA (<4.5/5) groups to compare their habits	The study found that self-fulfilment motivation, active recall, and a preference for visual learning were significantly associated with higher GPAs. High-achieving students also tended to attend lectures, study in the morning, prepare before class, make weekly schedules, and revise lectures. However, no strong link was found between GPA and habits like studying alone, regular reviewing, or writing out memorized content.	The study is limited by its use of self-reported data, which may lead to recall or desirability bias, and its cross-sectional design, which only reveals correlations rather than causation. Additionally, focusing on a single institution within a specific cultural context reduces the generalizability of the findings. Using GPA as the sole measure of academic performance also overlooks other important aspects such as practical or clinical skills.
[5]	Impact of Time-Management on the Student's Academic Performance: A Cross-Sectional Study	Despite the known benefits of time management, its direct impact on academic performance among radiology students has not been studied. The study addresses how students' time-management behaviours influence their academic outcomes, especially in the context of the challenges brought on by online learning during the COVID-19 pandemic	**Design**: Cross-sectional, survey-based study **Sample**: 142 undergraduate radiology students at King Abdulaziz University (Jeddah, Saudi Arabia) **Instrument**: Self-designed 22-question Likert-scale questionnaire based on domains such as planning, preferences, and workload management **Analysis**: SPSS software (v25.0) using Pearson chi-square to assess the relationship between time-management behaviours and GPA. **GPA**: Categorized from administrative records (not self-declared)	The study found that students with higher GPAs (4.5–5.0) were significantly more likely to meet deadlines (69.2%) and use to-do lists or calendars (66.3%), indicating that these time-management behaviours positively impact academic performance. In contrast, 36.7% of students with lower GPAs (4.0–4.5) believed poor planning hurt their results. While most students agreed that preplanning was beneficial, less than half reported managing their time effectively. No strong link was found between GPA and broader planning habits like weekly or yearly scheduling. Overall, practical time-management habits—especially deadline discipline and task tracking—were associated with better academic outcomes.	The study's limitations include its cross-sectional design, which prevents establishing causality, and reliance on self-reported questionnaires that may introduce response bias. Its narrow focus on radiology students from a single university limits generalizability, while potential confounders such as sleep patterns and online learning disruptions could have influenced the results. The authors recommend conducting longitudinal studies and incorporating digital planning tools to gain deeper insights in future research.

[6]	Self-Regulated Learning Training Programs Enhance University Students' Academic Performance, Self-Regulated Learning Strategies, and Motivation: A Meta-Analysis	Self-regulated learning (SRL) is a critical issue of many university students who have no effective learning strategies and managing their time use, which can be an obstacle to academic progress and motivation. Existing meta-analyses targeted school children or exceptional university students, and there were no meta-analyses to determine the effectiveness of SRL training programs targeting typical university populations and the effects of such trainings on specific learning strategies and motivational factors	**Design:** Meta-analysis (systematic review and statistical synthesis) **Sample:** 49 studies, 55 samples, totalling 5,786 university students **Method:** Included only extended SRL training programs for university students with direct instruction of learning strategies, experimental or quasi-experimental pre-post control group designs, and outcomes such as academic performance, learning strategies, or motivation. **Analysis:** Three-level meta-analysis of 251 effect sizes; moderator analyses to examine effects of training design (feedback, cooperative learning, learning protocols), theoretical background, and student characteristics (age, prior achievement) on outcomes	In general, meta-analysis revealed that self-regulated learning (SRL) training programs had a modest yet significant impact on the academic performance, metacognitive and resource management strategies, and motivation of university students. The most significant improvements were observed on metacognition and resource management skills, particularly those trainings incorporating either feedback, cooperative learning or learning protocols. These outcomes were observed in students of all grade and ability levels, although improvement in GPA may not be noticeable as soon.	Findings are based predominantly on self-report instruments and indicated variety in study design and quality. There was some publication bias, and the majority of the studies did not report on long-term effects or give a detailed description of the training, which restricts the generalizability of the results.
[7]	Factors affecting students' performance on national assessments of mathematics in Italy: A random forest approach	The study identifies key internal and external variables that affect student's math test scores in Italy.	**Design:** Quantitative, predictive modelling **Sample:** 300 students **Model:** Random Forest, Boosting Regression, Regression Tree **Metrics:** MAE, MSE, RMSE, R-square	Boys consistently achieve higher scores on the INVALSI test across all grades and more significant when the grade level increase. Besides that, student in Northern Italy have the highest average scores compared to the other.	Socioeconomic status excluded due to data gaps and the dataset only contain one year student analysis.
[8]	Key factors influencing students' academic performance	To identify which student-level factors significantly affect academic performance.	**Design:** Predictive regression study **Sample:** 10000 students **Model:** Linear Regression **Metrics:** R-square. MSE, MAE	The model showed that study hours, use of past exam questions, and previous academic performance were the most significant internal factors. External factors like participation in extracurricular activities also positively affected grades. The model achieved a high explanatory power with an R^2 of 0.98.	No RMSE were reported which limits understanding of prediction accuracy. Besides that, he high R^2 suggests potential overfitting.
[9]	Determining Factors That Affect Student Performance Using Various Machine Learning Methods	The study sought to identify which factors are most strongly associated with the final academic grade (G3) of Portuguese high-school students, and to validate their predictive impact using machine learning models.	**Design:** quantitative, observational design. **Sample:** 395 secondary school students and 33 variables **Model:** Pearson correlation, multilayer Perceptron, Decision tree, and Random Forest. **Metrics:** Pearson correlation coefficient (r), RMSE	Past academic failures are the strongest negative predictor of final grade. Besides that, the student has better academic performance when their mother education is higher.	The sample size is small compared to 33 attributes, posing to risk of overfitting

[10]	Using Decision Trees and Random Forest Algorithms to Predict and Determine Factors Contributing to First-Year University Students' learning Performance	This study aims to build a predictive model to estimate the learning ability of first-year university students.	**Design:** comparative, predictive modelling study **Sample:** 2407 first-year student from 12 departments at a vocational university in Taiwan **Model:** Decision Trees and Random Forests **Metrics:** Accuracy, Standard deviation, and Feature importance (Gini index)	The CART decision tree performed better in analysing. Both models provide explainable insights for strategic academic intervention	Limited to one vocational university in Taiwan might lead to biased results
[11]	Classification and Prediction of Student Performance Data Using Various Machine Learning Algorithms	This study compares the accuracy and predictive power of various machine learning algorithms in predicting students' academic performance	**Design:** experimental classification study **Sample:** 649 instances and 33 attributes from the UCI student performance dataset **Model:** SVM, Naïve Bayes, ID3, Decision Tree **Metrics:** Accuracy and Error rate	SVM achieved the highest accuracy among the compared algorithms	Focused more on comparative model accuracy rather than deep variable influence
[12]	Factor that Affects the Students' Performance in Mathematics Subject Using Logistic Regression	This study aims to identify factors affecting students' performance in MAT133 using regression model	**Design:** explanatory, statistical modeling approach **Sample:** 122 students of Diploma in Computer Sciences (CS110) in UiTM Cawangan Melaka Kampus Jasin **Model:** Binary Logistic Regression **Metrics:** p-values, Wald statistics, and Odds Ratios	The regression model was statistically significant and could be used to predict pass or fail likelihood in MAT133	Limited to a small sample size which only 122 samples are chosen and focused only on a single course and institution might lead to biased result

C. Writing the Literature Review

1) External factors that influence student performance

The first three articles collectively demonstrate that external factors play a significant role in the academic performance of students. The first article focusses on parental involvement, found that active engagement from parents such as reading together with children, monitoring homework and communication with teachers have a positive correlates with higher academic achievement, although the effect size varies according to socioeconomic status [1]. The second article examined the influence of family socioeconomic status and showed that students from higher-SES households perform better academically due to greater access to educational resources, superior parental expectations and more financial investment in schooling [2]. The third article examined school environmental conditions and found that well-ventilated, well-lit classrooms with adequate instructional materials are associated with improved concentration and higher test scores, whereas poor physical conditions negatively affect academic performance [3]. However, these articles share similar limitation, which is they examines these factors separately, relies on cross-sectional data and is geographically restricted ([1] U.S., [2] China, [3] Nigeria), limiting generalizability. Additionally, [1] and [2] use self-reported data, while [3] focuses on a single subject area.

2) Internal factors that influence student performance

According to several research papers, internal factors, such as study habits, time management, and self-regulation play a major role in determining the performance of the students. Studies among medical students indicate that self-discipline, proper study habits, and frequent academic reviews are influencing factors positively associated with academic success, but the association may be contextual and self-reported data [4]. In other research, the importance of good use of time, including preplanning, meeting deadlines, and using to-do lists, has been emphasized and demonstrated to enhance grades and aid in managing stress and avoiding procrastination, which are closely related to student mental health [5]. Furthermore, efforts aimed at developing self-regulated learning skills over cognitive, metacognitive, and motivational approaches indicate significant increases in academic achievements and resilience, which promotes academic task challenges that students have to bear [6]. All these studies have a common thread of similarity that is central in academic and psychological well-being is self-guided inner behaviours [4][5][6]. Nonetheless, one critical research gap is the overwhelming body of research that tends to consider these factors separately, instead of co-occurring, suggesting the need to conduct more comprehensive and multi - faceted research on how study habits, time management, and self-regulation interrelate to influence student outcomes [4][5][6].

3) Evaluation metrics for the estimation model

The studies founded the importance of prior academic outcomes and family-related factors as key predictors of student performance. Doz and Bregant [7] applied Random Forest and Boosting Regression to Italian INVALSI mathematics examination outcomes and certified grade as the best predictor, followed by gender and territorial area effect. Suleiman et al. [8] used multiple linear regression on a comprehensive student dataset and achieved an R^2 of 0.98, demonstrating that study habits, past performance, and socioeconomic background collectively explain nearly all variance in academic scores. Beckham et al. [9] conducted Pearson correlation screening and trained multilayer perceptron, Random Forest, and decision tree models on Portuguese secondary school students. They determined that past class failures had a strong negative correlation with final grade ($r \approx -0.36$), mother's education had a moderate positive correlation ($r \approx +0.22$), and that the multilayer perceptron model produced the lowest RMSE of 4.32 among the tested algorithms.

4) Type of predicting model

In recent years, educational institutions have increasingly adopted predictive models to forecast student academic outcomes. The estimation results can be applied in various sectors, such as identifying at-risk learners [10] and supporting scholarship distributions. The diversity of available predictive techniques provides various levels of accuracy, interpretability, and adaptability. According to the current findings [10][11][12], we found that decision trees, random forests, SVM, and regression analysis are the most commonly used predictive models due to their user-friendly and accuracy characteristics.

Huynh-Cam et al. successfully leverage 18 demographic and family background variables to predict student performance before they enter university [10]. This study illustrates the utility of tree-based models for early intervention. The decision tree model benefits from its rule-based interpretability, while the random forest model enhances prediction stability and reduces overfitting through ensemble averaging. Next, Pallathadka et al. found that Support Vector Machines (SVM) have higher accuracy than Naïve Bayes, ID3, and C4.5 [11]. This is due to its ability to handle high-dimensional data and complex decision boundaries. In addition, regression is also often used by researchers due to its statistical rigor, interpretability through odds ratios, and ability to assess both categorical and continuous predictions. While a regression model might lack the complexity to capture nonlinear relationships compared to others, it remains a robust tool for explanatory modelling in educational research [12].

D. Summary of Literature Review

In summary, this literature review shows a wide range of research that categorized into four key areas: internal factors, external factors, matrix frameworks and model-based approaches. It demonstrates that both students' personal behaviours and environmental conditions have a significantly influence on students' academic outcomes. While internal factors such as learning habits and motivation are more direct to student agency, external factors such as socioeconomic background and school environment impact more onto the opportunities for success. Matrix and model-based articles give essential theoretical and structural insights, helps in the organization and analysis of complex educational variables, as observe from the above section, each model has its own strengths and limitations. Overall, the reviewed literature provides a balanced view, limitations in methodology, geographical scope and integration across factors suggest a need for more comprehensive and inclusive research in the future.

III. Methodology

A. Introduction to the Methodology

Throughout the research, linear regression, decision tree, random forest and support vector machine (SVM) are used as the technique for our methodology. Each method is chosen for its unique strengths in handling different aspects of predictive modelling. For linear regression, it helps on interpretability to examinate the coefficients to see which features have stronger influence. For decision tree, it shows a visual structure of the feature selected were use for splitting and how often. For Random Forest and SVM, it helps on handling complex non-linear relationships. By comparing these models, the study aims to identify which factors have a greater influence on predicting students' academic performance based on personal or external factors.

B. Flow Diagram or Research Framework/Architecture

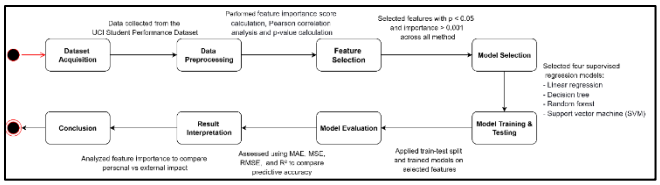


Figure 1: Flow Diagram

C. Research Design

The research uses quantitative approach for its involvement in numerical data, statistical model, performance metrics and a mathematical evaluation to answer research question, with a comparative study design focus on the predictive performance of multiple machine learning regression models. Where the goal is to evaluate and compare the influences of the four different models: Linear Regression, Decision Tree, Random Forest and Support Vector Machine (SVM), in predicting students' final grades (G3) based on selected personal and external attributes.

D. Steps Involved in the Methodology

The primary focus of this research is to investigate the impact of personal and external factors on students' academic performance and determine which category contributes more significantly to the final Math grade (G3) by using four different models. Additionally, the research aims to evaluate and compare the predictive accuracy of these four machine learning models and identify the most suitable model to be used for this educational performance.

1) Literature Review and Analysis

The existing solutions showcased in this study were gathered using a systematic review of relevant literature sourced from reputable academic databases, including Google Scholar, ScienceDirect, and Scopus. The words student academic performance, external factors, matrix frameworks, and model-based approaches were used as keywords in identifying appropriate articles. Inclusion criteria focused on peer-reviewed articles published within a decade that provided empirical or conceptual insights into factors influencing academic performance. Studies with unclear methodology or opinion-based were excluded and not directly related to the educational context. From this body of literature, common methods such as the use of parental involvement programs, improvements in school infrastructure, and the use of analytical models were discovered. The solution presented in this study integrates both internal and external factors into a unified conceptual framework, which addresses a gap in existing literature where these elements are often examined in isolation. This integrated framework is novel because it considers effects of multiple factors and incorporates structured analytical tools to capture the complex factors better that influence academic success.

2) Criteria for Comparison

Four regression performance metrics were used to evaluate the models to ensure a fair and balance comparison, The four metrics were:

- **Mean Absolute Error (MAE):** Measures the average magnitude of prediction errors between the paired observations.
- **Mean Squared Error (MSE):** Represent the average squared differences between predicted and actual values.
- **Root Mean Squared Error (RMSE):** Evaluate the by calculating error magnitude in the same unit as the target variable.
- **R-squared (R^2):** Indicates how well the model explains the variance in the final grade.

These metrics are used to assess the accuracy, robustness and clarification of each model. In addition of using these metrics, feature importance analysis is also conducted to identify whether personal or external factors contributes more significantly into the model predictions.

Data Collection

Variable Name	Type	Description
sex	Binary	Student's gender where 'F' is female, and 'M' is male
age	Integer	Student's age from 15 to 22
Medu	Integer	Student's mother education where '0' for none, '1' for primary education (4th grade), , '2' for 5th to 9th grade, '3' for secondary student and '4' for higher education
reason	Categorical	Reason to choose this school
traveltime	Integer	Home to school travel time where '1' for lesser than 15 minutes, '2' for 15 to 30 minutes, '3' for 30 minutes to 1 hour and '4'for more than 1 hour.
failures	Integer	Number of past class failure where '0' for no failures, '1' for failed 1 class, '2' for failed 2 classes, '4' for 3 or more failures
schoolsup	Binary	'Yes' for having educational support, 'No' for not having educational support
romantic	Binary	'Yes' for having romantic relationship, 'No' for not having romantic relationship
absences	Integer	Number of school absences for student from 0 to 93
G1	Integer	First period grade from 0 to 20
G2	Integer	Second period grade from 0 to 20

Figure 2: Data Dictionary

We had used a student performance dataset from UC Irvine Machine Learning Repository [13]. This dataset was collected from student reports and questionnaires based on mathematic subject. There have 395 observations for us to compared. From 33 attribute, we have chosen 'sex', 'age', 'Medu', 'reason', 'traveltime', 'failures', 'schoolsup', 'romantic', 'absences', 'G1', 'G2' as our focus attributes to compare. These attributes were chosen to analyze how internal and external factors affect academic performance. We compared differences in G1 and G2 to assess the impact of these factors on students' academic progress throughout the school year.

3) Experimental Setup

The experimental setup for this research was implemented using Python as it supports data analysis and visualization. The dataset was first loaded using the pandas library. The objective of this study is to predict the final grade (G3) based on students' earlier academic performance (G1 and G2) as well as selected internal and external factors. Exploratory data analysis (EDA) was conducted to visualize how each factor affects G1 and G2. Statistical test such as t-tests and ANOVA was used to compare different in academic performance across categories of each factor. The analysis also aimed to compare the effect of internal and external factors. In addition, grouped averages of G1 and G2 were also calculated across different factor categories to observe general performance patterns. This helped assess whether certain personal or external factor are associated with better or worse academic performance over the school year. Moreover, all visualizations were generated using matplotlib and seaborn. This supported the development and evaluation of regression models designed to predict G3, and enabled a clear comparison of how internal and external factors influence students' academic performance

4) Data preprocessing

Before conducting data analysis, data preprocessing had been carried out to ensure the accuracy and reliability of the dataset. For example, feature selection had been performed by using Python to reduce model complexity, avoid overfitting, and improve interpretability. The feature selection technique is crucial for selecting only the most relevant attributes that significantly affect the target variable, G3 in our analysis. Three feature selection methodologies had been developed, including feature importance score calculation and p-value calculation. Only the attributes that fulfil all three feature selection criteria will be selected as the attributes in our final analysis. The selected attributes need to fulfil the p-value less than 0.05 and feature importance score more than 0.001. As a result, 11 out of the 33 original attributes were retained for further modelling and analysis as stated above.

5) Implementation of Solutions

After analysing all the methodologies that had been taught throughout this semester and the ones stated in the academic journals, our group finally came out with four models in our analysis which are linear regression, decision tree, random forest, and support vector machine (SVM). The objective of the analysis is to find out the best model to predict the final grade (G3) of students based on selected input attributes. Four of the above estimation models are chosen as the target is a numerical variable that shows the final grade of the students in the Math subject.

Firstly, the linear regression model is a parametric model that assumes a linear relationship between the dependent variable and the independent variables. The model will then fit a linear equation of:

$$G3 = \beta 0 + \beta 1X1 + \beta 2X2 + \ldots \beta nXn + \infty$$

The linear regression model uses the least squares method to minimize the differences between the actual and predicted values.

Secondly, the decision tree regressor model is a rule-based non-parametric model that splits the dataset into branches based on conditions on feature values. It will be performed in a tree structure, so it is easy to understand and visualize. The algorithm recursively splits the data by selecting different attributes and thresholds that end up with homogeneous leaf nodes.

Thirdly, the random forest regressor model is an ensemble learning method that builds multiple decision trees and aggregates their outputs to enhance accuracy and stability. With the help of the bootstrapping technique, multiple sub-datasets will be generated for each of the decision trees, which prevents the bias and skewness of the original dataset. The prediction is then made by averaging the outputs of all trees to avoid overreliance on a single prediction.

Lastly, the SVM is a supervised machine learning algorithm that creates a model by finding the optimal separating hyperplane, which generalizes well. It maximizes the distance between the closet data points from different classes, ensures that the model remains robust to unseen data and generates reliable output. It is designed to fit the best line that falls within a margin of tolerance to predict a continuous output.

6) Evaluation and comparison

Among the four models that are included in the analysis, there are some similarities and differences between all of them. For the similarity side, all four models fall under the category of supervised learning algorithms. This means they are trained using a dataset where the input features are paired with a specific known target. All the models are trained to forecast the students' Math academic performance (G3) based on the given inputs. Unlike unsupervised learning algorithms, a clear targeted output is labelled, and the models do not need to group the data according to hidden patterns and structures. Moreover, all four models are commonly used in predictive analytics. Since the target variable is a continuous numerical variable in the range from 0 to 20, the most famous predictive analytics models are selected to predict the output. Furthermore, all four models applied using similar regression performance metrics, which are Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE), and R-squared (R2). These metrics help assess how well the models generalize to unseen data and allow for fair comparison among methods.

One of the most obvious differences is the model types. Linear regression model is a parametric model, while the other models are non-parametric models. The parametric model makes a strong assumption about the form of the linear relationship and tries to estimate a fixed number of parameters. The non-parametric model does not have a predefined form, which varies in every analysis based on the dataset and attributes. In addition, all four models have different robustness to noise. Linear regression and decision trees are vulnerable to noise as the models are designed for a simple analysis process. The preprocessing process is essential for them, as the bias of input might have a significant impact on the final output. The random forest and SVM have a better way to handle biased data. For example, the bootstrapping and ensemble averaging of the random forest generate the final output through averaging multiple trees that can prevent over-reliance on a single dataset.

7) Validation of result

The findings were validated by comparing them with previous published benchmarks, which are Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE), and R-squared (R2). The objective of this benchmark comparison is to evaluate and compare the predictive performance of the four selected models using proven standardized performance metrics. The MAE measures the average absolute difference between the predicted and actual value of G3. It is commonly used to reflect the reliability of the model by providing the magnitude of error in the same unit as the target variable. Next, MSE and

RMSE act as the average square differences and square roots of them respectively. These two matrices play a similar role as MAE to find out the differences between predicted and actual values. The lower the value indicates the better the predictive performance. Moreover, R2 is often used to represent the proportion of variance in the target variable that the model explains. The closer the R2 to 1, the better the model explains the target variable. A good explanation of the target variable will then result in higher accuracy of the model. The benchmark comparison using MAE and R2 clearly highlights the strengths and weaknesses of each model. These metrics provided a quantitative basis for selecting the most suitable model for predicting the students' final performance in Math subjects.

E. Tools and Technologies Used

Python was chosen as the primary programming language for this project due to its powerful ecosystem of libraries, ease of use, and flexibility in handling the full data science pipeline. Python provides comprehensive libraries for implementing all four estimation models used in this project. By using the scikit-learn library, all four models can be simply applied with only a few key instructions. Other than that, Python provides a strong visualization tool that is crucial for exploratory data analysis and communicating results. The matplotlib library can perform a wide range of static and interactive plots, especially the bar chart that used to compare the accuracy among the models.

F. Challenges and Limitations

One of the challenges during this project was the limited dataset size where only contain 395 data. This small sample size cannot represent overall performance patterns to all students. A s a result, the findings may not use to generalize all students. Another limitation is the possibility bias in self-reported data. For example, the number of absences or the reason for choosing a particular school may be subjective or, which can affect the reliability of the analysis. Furthermore, the dataset only collects Math subject scores, which cannot be generalized to other subjects, such as language or science.

G. Summary

This methodology had outlined the complete procedure to predict the student's final math grade (G3) using the previous academic performance from G1 and G2 and various internal and external factors. The process began with data collection from the UC Irvine Machine Learning Repository. After that, data preprocessing was carried out to ensure the model accuracy and effectiveness. One of the key parts during the process is feature selection where we had implemented using Python. Three techniques were used during the feature selection which is Pearson correlation, p-value and feature importance scores. Besides that, exploratory data analysis and statistical testing helped to visualise the relationships and support model assumption. Finally, four regression models (linear regression, decision tree, random forest and support vector machine) were trained and evaluated to find the best approach. By using this methodological step, we fulfil the core objectives where we need to predict which factors will affect the student academic performance.

IV. Results And Discussion

A. Introduction to Results

The below section shows the results of the four models our group used in predicting the students' final grade in the Math subject. The one with the highest accuracy and lowest error will be chosen as our final model in this project. After we have a deep learning of each model, we performed them and output their MAE, MSE, RMSE, and R^2. We found out that the decision tree regressor showed the best results among the four models, and we chose it to be our final predicting model for student performance according to personal feature and contextual feature.

B. Presenting Results

Machine learning technique	**MAE**	**MSE**	**RMSE**	**R^2**
Linear regression	1.1757	2.9848	1.7276	0.8028
Random forest	0.9710	1.6390	1.2803	0.8917
Decision tree	0.8148	1.2963	1.1386	0.9144
SVM	1.3386	4.5911	2.1427	0.6967

Figure 3: Result

As shown in the table above, decision tree regressor is the best model for predicting the students' final grade in the Math subject. Random forest regressor will be the second option as it also provides a high accuracy that is slightly lower than the decision tree and followed by linear regression models. SVM provides the lowest accuracy among the four models. This might be due to the dataset size is too small, as it is normally used for a more complex analysis situation.

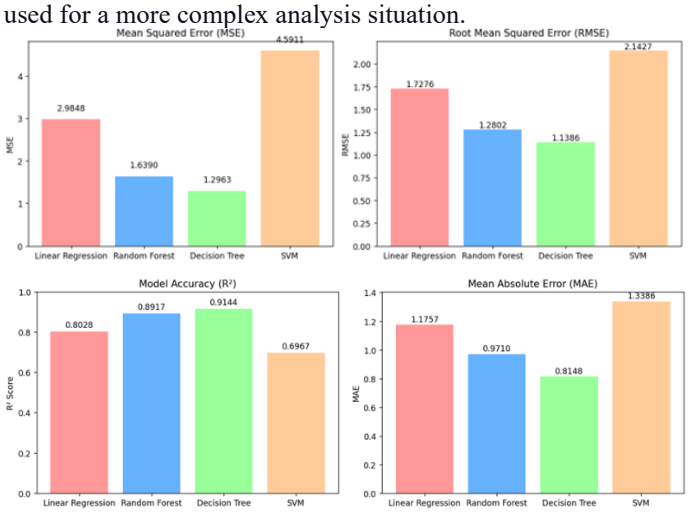


Figure 4: Bar chart for MAE, MSE, RMSE, and for R^2 each model

C. Comparative Analysis and Decision

As the bar charts above show, the four-evaluation metrics were applied by our group, including MAE, MSE, RMSE, and R^2. We compare the different predicting models from two

directions, which are error-based metrics and the proportion of variance explained. The error-based metrics include MAE, MSE, and RMSE that analyse the distance between the predicted and actual value of the target variables. A lower error means the model is making fewer mistakes and generating a reliable output. Moreover, the variance-based metric covers the $R^{2,}$ which focuses on how well the model explains the variability in the outcome. It reflects the model interpretability of the overall structure and relationship in the data. A high R^2 indicates the model can capture the meaningful trends and discover the majority of the significant variables that influence the target variable.

Among the four selected prediction models, the decision tree shows the best result in both error-based and variance-based metrics. The decision tree model receives the lowest number of errors in all three error-based metrics. It gets 0.8148,1.2963, and 1.1386 in MAE, MSE, and RMSE respectively as shown in the bar charts above. This reflects that the decision tree has a high accuracy in predicting the students' final grade, especially compared to SVM, which has MAE three times higher than its. Besides, the decision tree model also provides the highest R^2 score, which is 0.9144. This shows that 91.44% of the variability of the students' final grade has already been included through the decision tree model. The model had explained most of the significant variables that indicate that the result of the decision tree model is highly reliable and trustworthy.

D. Overall Discussion

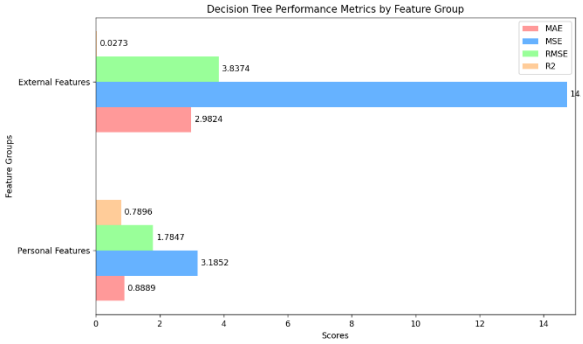


Figure 5: Decision Tree Performance Metrics by feature group

After the best-predicting model, the decision tree model was chosen, which is used to perform the analysis for comparing personal and external factors that influence students' final performance in the Math subject (G3). The selected significant features are divided into two sets. The personal attributes include sex, age, number of past failures, romantic relationship status, school absences, and previous grades G1 and G2. Besides, external attributes include the mother's education level, reason for choosing the school, travel time, and access to school support services. The objective of this study was to analyse whether student performance in school is more likely to be related to personal variables or influenced by external factors. The predictive performance of both feature sets was assessed using the four-evaluation metrics stated above, which are MAE, MSE, RMSE, and R^2. Both MAE, MSE, and RMSE metrics are used to understand the presence of errors from the personal and external groups, while the R^2 is used to predict the variability of the factors toward the decision tree model.

The clustered bar chart above shows the result of the analysis. The bar chart successfully reflects that the personal attributes demonstrated significantly stronger predictive power compared to the external attributes.

	Personal features	External features
MAE	0.8889	2.9824
MSE	3.1852	14.7259
RMSE	1.7847	3.8374
R^2	0.7896	0.0273

Figure 6: Result of the analysis

From the error-based metrics, the personal features have all MAE, MSE, and RMSE lower than the external features. This reflects that the result of the personal features ends up with more accurate and fewer differences from the actual results. From the variance-based metrics, the personal features also provide a higher significant level compared to the external features. The R^2 of the personal features is 0.7896, while the external features only 0.0273. This indicates that the personal attributes can influence 78.96% of the result of the decision tree model for the student performance analysis. As a result, we conclude that the student's performance in the Math subject is more related to their personal factors rather than the external influences.

The finding strongly supports the personal characteristics of the students, which are sex, age, number of past failures, romantic relationship status, school absences, and previous grades G1 and G2, play a dominant role in determining a student's final performance in the Math subject. This is because the age, number of past failures, G1, and G2 reflect the cumulative nature of learning in math, where foundational understanding heavily impacts future success. The student who receives a lesser past failure and higher G1 and G2 scores is most likely to get a better result in the final exam. Next, the school absences and the romantic relationship status reflect whether the student is attending and listening in the class. This is essential for students to catch up on the topic and know the techniques to answer the questions that might come out during the exam. Moreover, the male students are more likely to get a better result in the Math subject. This is because most of the male students have a higher interest in the Math subject when compared to the female students.

There are also some limitations in our analysis. We only focus on the Math subject might lead to different results when compared to the student performance from a wider subject range. For example, the sex factors might affect the students' performance in the Math subject, but this situation might be reversed in the design-related subjects. Future research should collect the data from a wider angle, such as calculating the average scores of each student instead of a single subject. Next, the current dataset only includes samples from two schools. This might lead to the bias of the school's professionals for only a certain subject. The future research should gather data across as many schools as possible to reduce the risk of biased data.

E. Critical Evaluation

The main objective of this study was to develop a reliable model for predicting students' final math grade (G3) using earlier performance (G1 and G2) and internal and external factor attributes. Based on the model performance evaluation, the decision tree provides the best result, where the MAE is only 0.8148 (lowest) with r-square 0.9144 (highest) among other models. This indicates the model able capture complex patterns and explain over 91% of the variance in the G3 result.

Additionally, the study found that the personal factors had much significant predictive influence on G3 compared to external factors. Their G1, G2, past failures, absences and romantic status shows higher accuracy and consistency across all error-based and variance-based metrics. These finding highlight the importance of past academic performance and personal factor in predicting their future academic results.

F. Linking to Literature Review

The findings of this study align closely with existing literature, confirming the importance of both external and internal factors in influencing students' academic performance. Consistent with [1][2][3], the findings showed that students with higher levels of parental involvement, greater family socioeconomic status, and access to better school environmental conditions achieved higher academic scores, with greatest improvements language subjects. For instance, the study's observation that students from well-resourced schools outperformed students from limited recourse environments mirrors the environmental quality effects reported in [3]. Similarly, the positive relationship identified between parental engagement and student academic performance supports the conclusions of [1], whereas the mediating role of socioeconomic resources affirms findings in [2]. The use of structured categorization in this study also reflects the matrix-based approaches discussed in [7][8][9], enabling clearer identification of interacting variables. Furthermore, the integrated conceptual framework applied here builds on theoretical models from [10][11][12], combining multiple dimensions of influence rather than examining them in isolation. No major inconsistencies with past research were uncovered; rather, this research expands on past research by demonstrating how exogenous variables can be sensibly organized and tested in order to gain a better understanding of their overall effect on academic performance.

G. Conclude the Results and Discussion

In conclusion, this research successfully used four machine learning models to predict the final Math grade (G3). We used a combination of early academic results with internal and external factors. The decision tree model provided better result than others based on MAE, MSE, RMSE, and R^2. Thus, we have chosen it as the final model for further analysis. Furthermore, the comparison between internal and external features clearly demonstrated that personal factors had significantly higher predictive accuracy compared to external factors. Personal factors like past performance (G1 and G2), absences, past failures, and romantic status. Thus, the study had found the significance of early academic performance and student behavior in predicting outcomes, providing insights that can support targeted interventions. We recommend the future work could extend this research by including more subjects or broader datasets to improve generalizability.

V. Conclusion And Future Work

The purpose of the study was to compare the relative predictive ability of personal and external variables in predicting final mathematics performance of students through the use of a more advanced regression model and a strict feature selection methodology. It is evident that over all the stages and results, the decision tree model performed remarkably better than the other three approaches and produced the highest predictive accuracy as seen by R^2 being approximately 0.91. More to the point, the analysis revealed that individual factors such as previous grades, attendance record, and track record of academic failures were significantly more important determinants of academic outcomes than external factors like parental education or school support. The results directly refer to the purpose of the study, as they also reveal that the internal characteristics of students play a pivotal role in predicting effective grades.

This study contributes to the area of educational data mining by empirically estimating the relative role of various categories of factors. The three main contributions are: an empirical distinction between various machine learning models of predicting grades, straightforward evidence of the relative importance of personal factors in determining student success relative to external ones, and the feasible evidence to justify early emerging and intercession strategies on at-risk students. The reflections here serve as great insight to provide information to educators and policymakers to optimize resource allocation and personalizing support services.

Although the results are solid considering the framework of the data and procedures employed, a number of restrictions should be mentioned. The data used to conduct this analysis only included grades in mathematics over a specific and relatively small group of students within two schools. Therefore, these findings might not be entirely generalizable to other people, academic classes, or the overall world. Furthermore, the study failed to consider other possible unmeasured factors including student motivation, mental welfare, or peer influence, which can also be important.

In future studies, the limitations of small school and student sample sizes as well as narrow range of subjects and school types should be resolved by including larger and more varied groups of students in various schools, subjects and regions. It can be further improved by increasing the size of the feature space to cover socio-economic indicators, psychological attributes, or even behavioural data, to boost predictive results. The challenge of testing advanced ensemble or deep learning models, as well as the implementation of the models into functional early-warning systems in school settings, are

areas of future motivation. It might also be interesting to observe the variance of predictive factor importance by subject domain or demographic of students and see what other actionable insights might be available.

In summary, the study indicates that predictive analytics is a great high holder in education. The study offers a starting point toward more individualized, data-based intervention plans by clarifying what influences student achievement the most: by clarifying which factors matter most to student achievement, the study offers a starting point toward more individualized, data-based intervention plans: helping schools and educators to proactively promote student success and ultimately enhance learning outcomes.

REFERENCES

[1] C. Y. Tan, M. Lyu, and B. Peng, "Academic benefits from parental involvement are stratified by parental socioeconomic status: A meta-analysis," Parenting, vol. 20, no. 4, pp. 1–47, Dec. 2020, doi: https://doi.org/10.1080/15295192.2019.1694836.

[2] H. O. Omolo, A. Otara, and K. B. Atieno, "School environmental factors influencing academic performance in secondary schools," *International Journal of Novel Research in Education and Learning*, vol. 7, no. 3, pp. 35–45, May–June 2020. [Online]. Available: https://www.noveltyjournals.com/upload/paper/School%20environmental%20factors-2354.pdf

[3] F. Zhang, Y. Jiang, H. Ming, Y. Ren, L. Wang, and S. Huang, "Family socio-economic status and children's academic achievement: The different roles of parental academic involvement and subjective social mobility," British Journal of Educational Psychology, vol. 90, no. 3, pp. 561–579, Aug. 2020, doi: https://doi.org/10.1111/bjep.12374

[4] M. A. Aljaffer et al., "The impact of study habits and personal factors on the academic achievement performances of medical students," BMC Medical Education, vol. 24, no. 1, Aug. 2024, doi: https://doi.org/10.1186/s12909-024-05889-y.

[5] A. Alyami, A. Abdulwahed, A. Azhar, A. Binsaddik, and S. M. Bafaraj, "Impact of Time-Management on the Student's Academic Performance: A Cross-Sectional Study," Creative Education, vol. 12, no. 3, pp. 471–485, Mar. 2021, doi: https://doi.org/10.4236/ce.2021.123033.

[6] M. Theobald, "Self-Regulated Learning Training Programs Enhance University Students' Academic Performance, Self-Regulated Learning Strategies, and Motivation: A Meta-Analysis," Contemporary Educational Psychology, vol. 66, no. 1, p. 101976, May 2021, doi: https://doi.org/10.1016/j.cedpsych.2021.101976.

[7] D. Doz and B. Bregant, "Factors affecting students' performance on national assessments of mathematics in Italy: a random forest approach," Assessment in Education: Principles, Policy & Practice, vol. 31, no. 5–6, pp. 325–352, Nov. 2024, doi: https://doi.org/10.1080/0969594x.2025.2457687.

[8] I. B. Suleiman, O. A. Okunade, E. G. Dada, and U. C. Ezeanya, "Key factors influencing students' academic performance," Journal of Electrical Systems and Information Technology, vol. 11, no. 1, Sep. 2024, doi: https://doi.org/10.1186/s43067-024-00166-w.

[9] N. R. Beckham, L. J. Akeh, G. N. P. Mitaart, and J. V. Moniaga, "Determining factors that affect student performance using various machine learning methods," Procedia Computer Science, vol. 216, pp. 597–603, 2023, doi: https://doi.org/10.1016/j.procs.2022.12.174.

[10] T.-T. Huynh-Cam, L.-S. Chen, and H. Le, "Using Decision Trees and Random Forest Algorithms to Predict and Determine Factors Contributing to First-Year University Students' Learning Performance," Algorithms, vol. 14, no. 11, p. 318, Oct. 2021, doi: https://doi.org/10.3390/a14110318.

[11] H. Pallathadka, A. Wenda, E. Ramirez-Asís, M. Asís-López, J. Flores-Albornoz, and K. Phasinam, "Classification and prediction of student performance data using various machine learning algorithms," Materials Today: Proceedings, Jul. 2021, doi: https://doi.org/10.1016/j.matpr.2021.07.382

[12] N. Mahmud, P. Nur, Rosli, Nur Syakirah, and Jamaluddin, Siti Hafawati, "Factor that affects the students performance in Mathematics subject using logistic regression / Norwaziah Mahmud ... [et al.] - UiTM Institutional Repository," Uitm.edu.my, Mar. 2024, doi: https://ir.uitm.edu.my/id/eprint/94349/1/94349.pdf

[13] P. Cortez, "UCI Machine Learning Repository," archive.ics.uci.edu, 2014. https://archive.ics.uci.edu/dataset/320/student+performance